

No.	Solution	Remarks
1(a)	Base units of ρ = Base units of $\frac{Mr^n}{L}$ $\text{kg m}^{-3} = \frac{\text{kg m}^n}{\text{m}}$ $\text{m}^{-3} = \text{m}^{n-1}$ $\Rightarrow n = -2$	[1] for correct workings
1(b)(i)	$\frac{\Delta r}{r} = \frac{1}{2} \left(\frac{\Delta \rho}{\rho} + \frac{\Delta M}{M} + \frac{\Delta L}{L} \right)$ $= \frac{1}{2} \left(\frac{50}{2340} + \frac{0.18}{100} + \frac{0.0002}{0.1242} \right)$ $= 0.012389$ ≈ 0.012	[1] for correct equation [1] for answer
1(b)(ii)	$\rho = \frac{Mr^{-2}}{kL}$ $2340 = \frac{(1.072)r^{-2}}{2.094(0.1242)}$ $r = 0.041970$ $\frac{\Delta r}{r} = 0.012389$ $\Delta r = 0.012389 \times 0.041970 \approx 0.0005$ $\therefore r = 0.0420 \pm 0.0005$	[1] for r [1] for correct expression
2(a)	Newton's third law of motion states that if body A exerts a force	[1]